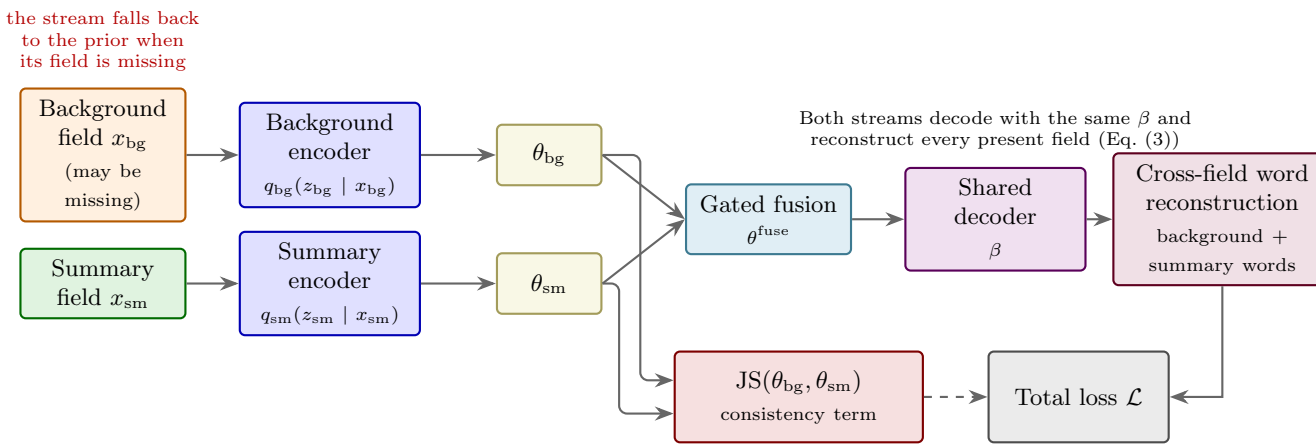

the stream falls back to the prior when its field is missing
Background field x_{bg} (may be missing)
Summary field x_{sm}
Background encoder $q_{\mathrm{bg}}(z_{\mathrm{bg}} \mid x_{\mathrm{bg}})$
Summary encoder $q_{\mathrm{sm}}(z_{\mathrm{sm}} \mid x_{\mathrm{sm}})$
θ_{bg}
θ_{sm}
Both streams decode with the same β and reconstruct every present field (Eq. (3))
Gated fusion θ^{fuse}
Shared decoder β
Cross-field word reconstruction background + summary words
$\mathrm{JS}(\theta_{\mathrm{bg}}, \theta_{\mathrm{sm}})$ consistency term
Total loss $\mathcal{L}$